9.PT-109 American patrol boat

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
    char key[] = "ROYALNEWZEALANDNAVY";
    char cipher[] =
    "KXJEYUREBEZWEHEWRYTUHEYFS"
    "KREHEGOYFIWTTTUOLKSYCAJPO"
    "BOTEIZONTXBYBNTGONE YCUZWR"
    "GDSONSXBOUYWRHEBAAHYUSEDQ";

    char matrix[5][5];
    int used[26] = {0};
    int i, j, k = 0, r1, c1, r2, c2;
    char ch;

    for (i = 0; key[i] != '\0'; i++)
    {
        ch = toupper(key[i]);
        if (ch == 'J')
            ch = 'I';

        if (ch >= 'A' && ch <= 'Z' && !used[ch - 'A'])
        {
            matrix[k / 5][k % 5] = ch;
            used[ch - 'A'] = 1;
            k++;
```

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    }
}

for (ch = 'A'; ch <= 'Z'; ch++)
{
    if (ch == 'J')
        continue;

    if (!used[ch - 'A'])
    {
        matrix[k / 5][k % 5] = ch;
        k++;
    }
}

printf("Playfair Matrix:\n");
for (i = 0; i < 5; i++)
{
    for (j = 0; j < 5; j++)
        printf("%c ", matrix[i][j]);
    printf("\n");
}

printf("\nDecrypted message:\n");

for (i = 0; cipher[i] != '\0'; i += 2)
{
    char a = toupper(cipher[i]);
    char b = toupper(cipher[i + 1]);
```

```
if (a == 'J') a = 'I';
if (b == 'J') b = 'I';

r1 = c1 = r2 = c2 = 0;

for (j = 0; j < 25; j++)
{
   if (matrix[j / 5][j % 5] == a)
   {
      r1 = j / 5;
      c1 = j % 5;
   }
   if (matrix[j / 5][j % 5] == b)
   {
      r2 = j / 5;
      c2 = j % 5;
   }
}

if (r1 == r2)
{
   c1 = (c1 + 4) % 5;
   c2 = (c2 + 4) % 5;
}
else if (c1 == c2)
{
   r1 = (r1 + 4) % 5;
   r2 = (r2 + 4) % 5;
}
else
```

```
        {
            int temp = c1;
            c1 = c2;
            c2 = temp;
        }

        printf("%c%c", matrix[r1][c1],
                        matrix[r2][c2]);
    }

    printf("\n");
    return 0;
}
```

Output:

